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Behavioral Modeling of Power Amplifiers Using Complex LUTs for OFDMA Signals via Cadence Spectre Simulation

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Abstract - This work develops and evaluates a Python-based simulation chain for digital predistortion of a Doherty power amplifier operating with OFDMA and QAM signals. The system generates multi-user complex baseband signals, converts them to passband, and applies them to a Cadence/Spectre circuit model. The amplifier is represented by a variable memory Lookup Table (LUT) model whose complex coefficients are obtained by nonlinear least squares, while a second model is trained with the inverse input-output relationship to perform digital predistortion. Training and validation signals are aligned and compensated for delay and complex gain before estimation. The method is evaluated using normalized mean squared error, bit error rate, error vector magnitude, AM-AM response, power spectral density, and constellation analysis. The resulting framework supports consistent comparison between behavioral and circuit-level simulations while reducing the complexity of nonlinear amplifier modeling.

I. Introduction

Modern telecommunications demand high data rates, making the Power Amplifier (PA) a critical component despite its nonlinear behavior and power consumption. The trade-off between linearity and efficiency is further affected by memory effects and circuit dynamics. Behavioral models provide a practical way to represent the PA input-output relationship and evaluate linearization techniques without repeatedly simulating the complete circuit. This is particularly relevant to OFDMA and QAM-based systems, which are widely used in high-throughput wireless communications [1], [2].

This approach reduces the computational cost of repeated transistor-level analyses while retaining the nonlinear behavior relevant to DPD. Cadence Spectre is used as an additional early-stage source of input/output data; the main contribution is the portable behavioral framework and its complexity–accuracy evaluation.

This work develops a Python-based simulation chain for digital pre-distortion of a power amplifier driven by multi-user OFDMA/QAM signals. The model is extracted and calibrated from input/output data acquired from a real GaN Class AB amplifier, and the same training flow can also be integrated directly with Cadence circuit simulations. The implemented model uses complex lookup tables (LUTs) with memory, nonlinear least-squares fitting, and automatic selection of the most relevant memory lines. A second LUT model is trained with the inverse PA relationship to generate the pre-distorted signal. Model accuracy and transmission quality are evaluated using normalized mean square error (NMSE), bit error rate (BER), error vector magnitude (EVM), AM-AM response, power spectral density, and constellation plots. The system scope is shown in Figure 1; antenna and RF-filter behavior are outside the scope of this implementation.

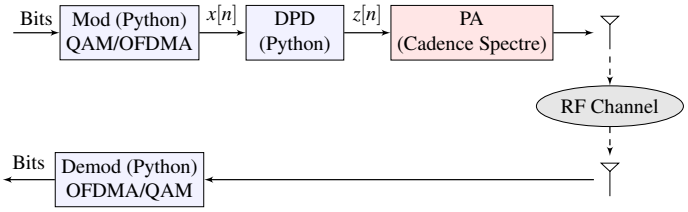


Figure 1: *Complete system block diagram from source bits to recovered bits. The RF channel is ideal AWGN, without fading.*

II. Mathematical modeling

The model parameters are estimated by least-squares optimization, minimizing the squared error between the measured PA output and the model estimate. The optimized parameters are complex LUT coefficients.

As a theoretical basis, the memory polynomial, a particular case of the Volterra series [3], is written as:

$$\tilde{y}(n) = \sum_{p=1}^{P} \sum_{m=0}^{M} \tilde{b}_{2p-1,m} \, |\tilde{x}(n-m)|^{2p-2} \, \tilde{x}(n-m) \tag{1}$$

where P and M are the maximum polynomial order and memory depth, respectively. This model motivates the nonlinear model with memory but is not directly optimized here.

The implemented model uses complex Lookup Tables (LUTs) to represent the nonlinear gain of each memory line. Five lines use 16, 12, 8, 4, and 2 interpolation points; the three most relevant are selected. For line m, linear interpolation gives [4]:

$$f_{\mathrm{LUT}_m}(|\tilde{x}(n-m)|) = \tilde{s}_m(q-1) + \left[\frac{\tilde{s}_m(q) - \tilde{s}_m(q-1)}{e_m(q) - e_m(q-1)}\right] \cdot (|\tilde{x}(n-m)| - e_m(q-1)) \tag{2}$$

where $e_m(q-1)$ and $e_m(q)$ bound the input amplitude and $\tilde{s}_m$ are complex LUT entries. The contribution of line m is

$$\hat{y}_m[n] = x[n-m] f_{\mathrm{LUT}_m}(|x[n-m]|). \tag{3}$$

The complete model is

$$\hat{y}[n] = \sum_{m \in \mathscr{A}} x[n-m] G_m(|x[n-m]|), \tag{4}$$

where $\mathscr{A}$ is the set of selected lines and $G_m(\cdot)$ is their complex LUT gain. The parameters are fitted by nonlinear least squares:

$$\min_{\{G_m\}} \sum_n \left| y[n] - \sum_{m\in\mathscr{A}} x[n-m] G_m(|x[n-m]|) \right|^2 . \tag{5}$$

The model is first fitted with all five lines. Each line is removed in turn, and the three causing the largest error increase are retained for a second fit. The line-specific grids are uniformly spaced from zero to the 99.9th percentile of the training amplitude.

Before fitting, the PA output is delay-aligned and corrected by a complex gain, then split into training and validation sets. For digital pre-distortion (DPD), a second LUT model maps the PA output back to the original input:

$$\hat{x}[n] = \sum_{m\in\mathscr{A}} y[n-m] H_m(|y[n-m]|), \tag{6}$$

where $H_m(\cdot)$ is the DPD LUT gain. The pre-distorted signal is passed through the PA model or Cadence simulation.

The transmitted signal uses Gray-coded 256-QAM in an OFDMA grid with four users, 12 contiguous subcarriers per user, and $N = 2048$. For $X[k] = I_k + jQ_k$, the baseband signal is

$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k] \exp\left(j\frac{2\pi kn}{N}\right) \tag{7}$$

where unassigned subcarriers are zero. The passband waveform is $s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t)$.

Performance is evaluated using BER, NMSE, and EVM.

The Bit Error Rate (BER) is the fraction of incorrectly received bits:

$$\mathrm{BER} = \frac{1}{N_2} \sum_{i=1}^{N_2} (b_{i,\mathrm{tx}} \neq b_{i,\mathrm{rx}}) \tag{8}$$

where N_2 is the number of compared bits. After gain and delay correction, received symbols are demapped to Gray-coded 256-QAM bits and compared with the transmitted sequence to obtain BER.

The Normalized Mean Square Error (NMSE) measures time-domain modeling accuracy in dB:

$$NMSE = 10\log_{10}\left[\frac{\sum_{n=1}^{N_1} |e(n)|^2}{\sum_{n=1}^{N_1} |y_{\mathrm{ref}}(n)|^2}\right] \tag{9}$$

where $e(n) = y_{\mathrm{ref}}(n) - \hat{y}(n)$ over N_1 validation samples. More negative values indicate smaller errors [5].

The Error Vector Magnitude (EVM) measures the deviation of received symbols from their ideal constellation positions and is used to evaluate the linearity of modulated power-amplifier signals [6]:

$$EVM(\%) = \sqrt{\frac{\frac{1}{N}\sum_{j=1}^{N} (\delta I_j^2 + \delta Q_j^2)}{S^2}} \times 100\% \tag{10}$$

where δI_j and δQ_j are the in-phase and quadrature errors, and S^2 is the average ideal constellation power. EVM is also reported in dB as $20\log_{10}(EVM)$ after complex-gain correction.

III. Python Codes

The Python implementation developed in this work integrates signal generation, Cadence/Spectre simulation, behavioral modeling, digital pre-distortion (DPD), and performance evaluation. The experiment is controlled by a main simulation script, with parameters grouped in `ExperimentConfig`.

The training and test signals are generated by `generate_OFDMA_signal()`. Random bits are mapped to QAM symbols, assigned to contiguous subcarriers for each user, and transformed to the time domain using the inverse discrete Fourier transform. The model can be extracted and calibrated from measured input/output data acquired from the real amplifier or trained from data obtained through direct Cadence integration. The complex baseband signal is converted to passband and exported as a PWL file when the Cadence circuit flow is used.

When enabled, `cadence_runner.py` transfers the `.scs` netlist and `.pwl` stimulus through SSH, adds the transient `tran` analysis, runs Spectre in batch mode, and downloads the traces. They are resampled, converted to complex baseband, aligned in time, and corrected for complex gain before training and validation.

For the dense 104-GHz PWL stimulus, `maxstep=1 ps` and `strobeperiod=1/F_s` balance transient resolution and solver cost.

The amplifier is modeled by `VariableLUTModel` in `lut_model.py`. Each LUT line represents a delayed sample, and its complex gain is obtained by interpolation as a function of the delayed-sample amplitude:

```python
def predict(self, input_signal, coefficients=None):
    coefficients = self.coefficients if coefficients is None else
    ↪  coefficients
    result = np.zeros(len(input_signal), dtype=complex)
    for line in self.config.active_lines:
        delayed = np.roll(input_signal, line)
        delayed[:line] = 0  # zero-pad boundary
        amplitude = np.abs(delayed)  # envelope
        grid = self.config.interpolation_grids[line]
        real_gain = np.interp(amplitude, grid,
        ↪  coefficients[line].real)
        imag_gain = np.interp(amplitude, grid,
        ↪  coefficients[line].imag)
        result += delayed * (real_gain + 1j * imag_gain)
    return result  # modeled output
```

Figure 2: *Prediction using the memory LUT model.*

The coefficients are fitted with SciPy's nonlinear least-squares solver [7], [8]. Since the optimizer operates on real vectors, the complex coefficients are packed and reconstructed afterward. `LUTLineSelector` evaluates memory-line contributions and retains the relevant lines. DPD training uses the inverse PA relation, with the amplifier output as input and the desired input as target.

After training, a new OFDMA signal is pre-distorted and passed through the PA model or a second Spectre simulation. The aligned output is used to calculate BER, EVM, and NMSE and to generate AM-AM, PSD, constellation, and time-domain plots, allowing behavioral and circuit-level results to use the same signal-processing chain.

The extraction and validation stages use the same post-processing chain. The complete implementation developed for this article is available at: github.com/andrepozzan/ic-gics

IV. Results

The signals analyzed in this work were obtained from a Class AB power amplifier produced with GaN technology. We employed a carrier centered at 900 MHz, modulated by an envelope signal suitable for WCDMA (with a bandwidth close to 3.84 MHz). Input and output waveforms were recorded using a Rohde & Schwarz FSQ vector signal analyzer, operating at a sampling rate of 61.44 MHz. The Class AB amplifier was used to learn the modeling method; the main Doherty results (AM-AM, constellation and power spectral density) use RF WebLab measurements [9], while the second Doherty is evaluated independently with Cadence Spectre data.

The signals were processed using separate extraction and validation datasets. Each dataset contains 9,995 complex input/output samples. The first set was used to identify the amplifier model and train the digital pre-distorter (DPD), while the second set was reserved for validation. Before fitting, the signals were aligned in time and corrected for their constant complex gain.

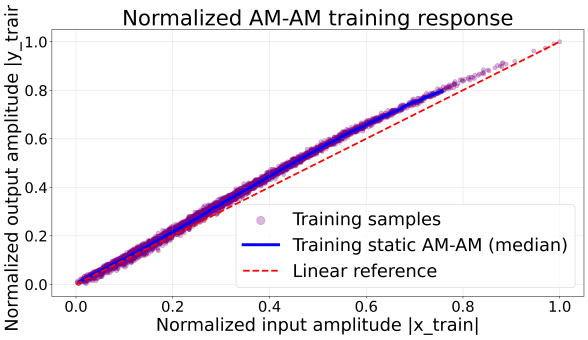


Figure 3: *Normalized AM–AM training response.*

The communication waveform uses ten users, twenty-four subcarriers per user, 256-QAM modulation, and an FFT size of 2048. The carrier frequency is 26 GHz, the sampling rate is 104 GSa/s, and a 6 dB input back-off is applied to keep the OFDMA waveform within the useful input range of the amplifier model.

For DPD training, the normalized AM–AM response used to fit the model is shown in Figure 3. The scatter points represent the training samples, the blue curve gives the static AM–AM response, and the red dashed line indicates the ideal linear reference. The close agreement confirms that the identified LUT captures the dominant static nonlinear behavior.

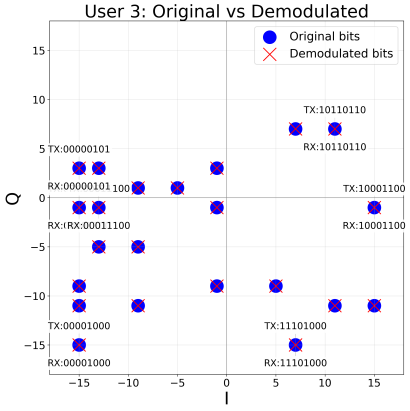


Figure 4: *Original and demodulated symbols for User 3 in the main simulation.*

The representative, error-free User 3 constellation (Figure 4) avoids visual clutter. The original symbols and demodulated symbols overlap after gain and phase correction, indicating that the DPD model preserves the QAM decisions.

The identified variable-memory LUT initially contains five memory lines and 84 complex coefficients. The line-selection procedure retained three lines, reducing the model to 72 coefficients. On the independent validation set, the model reached an NMSE of $-22.8370\,\mathrm{dB}$, as recorded in the execution log. For this behavioral amplifier, the communication metrics obtained after gain and phase correction are listed in Table 1. The effective SNR is estimated from the measured EVM according to $\mathrm{SNR}_{\mathrm{eff}} \approx -\mathrm{EVM}_{\mathrm{dB}}$; it therefore represents recovered-symbol quality, not an independent noise-power measurement; this approximation assumes an ideal AWGN-like channel and loses accuracy with severe phase noise or extreme nonlinear distortion. Users 0, 3, and 5 account for all 60 bit errors.

Table 1: Metrics for the behavioral amplifier in the main simulation.

BER	EVM	$\mathrm{SNR}_{\mathrm{eff}}$	NMSE
3.12%	0.3960%	48.05 dB	−22.8370 dB

The relevant output of the main simulation is reproduced below. It follows the same format as the program log and reports the BER for each user, the EVM, and the validation NMSE.

```
Selected memory lines: (0, 1, 2); coefficients: 84 -> 72
User 00: 15.62% (30/192 errors)
User 01: 0.00% (0/192 errors)
User 02: 7.29% (14/192 errors)
User 03: 0.00% (0/192 errors)
User 04: 0.00% (0/192 errors)
User 05: 8.33% (16/192 errors)
User 06: 0.00% (0/192 errors)
User 07: 0.00% (0/192 errors)
User 08: 0.00% (0/192 errors)
User 09: 0.00% (0/192 errors)
Overall: 3.12% (60/1920 errors)
EVM: 0.3960% (-48.0454 dB)
```

Figure 5: *Metrics reported by the main simulation.*

The power spectral density comparison in Figure 6 shows the DPD output following the reference more closely than the uncompensated output; Figure 7 shows the corresponding time-domain agreement.

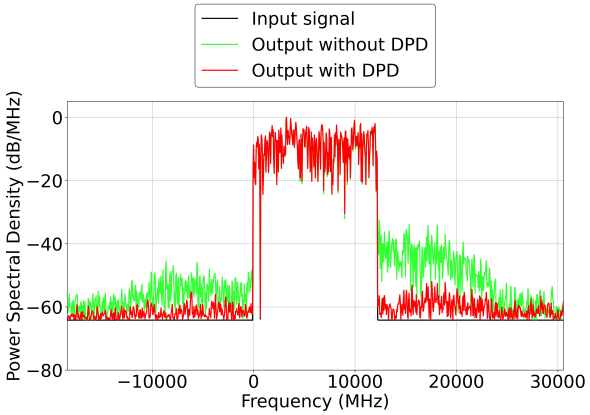


Figure 6: *Power spectral density of the reference and amplifier outputs with and without DPD.*

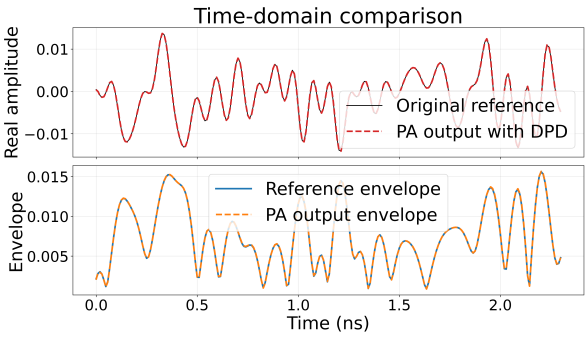


Figure 7: *Time-domain comparison between the reference waveform and the amplifier output with DPD.*

As an additional evaluation, a different Doherty amplifier from [6] was simulated at the transistor level in Cadence Spectre. This circuit is different from the behavioral amplifier, so its numerical results are reported independently. The testbench uses a 3.3 V supply, 2.8 V branch bias, 0.65/0.25 V gate biases, and 26 GHz at 104 GSa/s; topology details are given in the reference. The transient analysis saves the input and output nodes, which are converted to complex baseband and processed with the same demodulation and error calculations.

Table 2: Metrics for the Doherty amplifier evaluated with Cadence Spectre.

BER	EVM	SNR_{eff}	NMSE
7.03%	8.1187%	21.81 dB	−23.7596 dB

The 256-QAM flow used `LUTLineSelector` to remove redundant lines, reducing the least-squares matrix and coefficients from 84 to 44. It reported −23.759607 dB NMSE, 7.500 samples delay, and 27 errors in Users 1/2 (Table 2); SNR is derived from EVM rather than an independent noise-power measurement. The −23.76 dB result trades Volterra's greater precision and six additional iterations for the LUT's lower computational cost.

```
Selected memory lines: (0, 3, 4); coefficients: 84 -> 44
User 00: 0.00% (0/96 errors)
User 01: 11.46% (11/96 errors)
User 02: 16.67% (16/96 errors)
User 03: 0.00% (0/96 errors)
Overall: 7.03% (27/384 errors)
EVM: 8.1187% (-21.8103 dB)
Validation NMSE: -23.759607 dB
```

Figure 8: *Metrics of the Doherty Spectre execution.*

Therefore, the main simulation documents the extraction and validation of the behavioral amplifier model, while the Spectre results provide an additional, independent evaluation of the transistor-level Doherty amplifier. The two experiments are kept separate because they concern different amplifiers and therefore expose circuit-specific nonlinear behavior. Figure 8 also shows that the 27 errors are concentrated in Users 1 and 2, consistent with the aggregate BER reported in Table 2.

V. Conclusion

This research demonstrated a digital pre-distortion methodology based on a variable-memory lookup-table model for power-amplifier linearization. The behavioral extraction and validation flow showed that the DPD can improve the agreement between the amplifier output and the desired reference in the spectral, time-domain, and constellation analyses.

The complex-baseband framework is portable across carrier frequencies, requiring adapted sampling rate and input back-off. The Doherty Spectre test confirms transistor-level applicability, although results remain circuit-specific. Together, the experiments highlight the potential of the approach and the need to account for circuit-specific nonlinear behavior when moving from a behavioral model to a transistor-level implementation. Memory-line selection reduces the LUT search matrix and computational cost while retaining the behavior needed for DPD. Future work should investigate alternative circuit topologies, the influence of memory depth, and optimization algorithms that improve the trade-off between model complexity, convergence, and demodulation accuracy. These results support the use of behavioral models as a practical alternative to repeated transistor-level simulations during early amplifier development. The proposed flow preserves the communication-level evaluation while allowing the model complexity to be adjusted to the available computational budget.

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